

NAZIRU HALILU

UAV and Machine Learning Crop Phenotyping | Precision Agriculture
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[GitHub](#) | [Zenodo record list](#)

RESEARCH SUMMARY

Agricultural engineer specializing in UAV and sensor based high throughput phenotyping for cereal and row crops. Builds multispectral and hyperspectral imaging and vegetation index pipelines, self supervised and hybrid deep learning models (CNN, CNN-BiLSTM-Attention) for crop stress and yield trait detection, and reproducible, uncertainty aware machine learning workflows integrated with GIS, satellite remote sensing, and crop simulation models, directly transferable to image based, genomics informed plant breeding pipelines.

EDUCATION

Erasmus Mundus Joint Master's Degree (EMJMD), Sustainable Agriculture <i>Public University of Navarre, Spain / UTAD, Portugal / Agricultural University of Athens, Greece</i>	2025 to 2027
B.Eng., Agricultural and Bioresources Engineering <i>Ahmadu Bello University, Zaria, Nigeria. First Class Honours, CGPA 4.77/5.00, Top 1%</i>	2019 to 2024
National Diploma, Agricultural Engineering Technology <i>Division of Agricultural Colleges, ABU Zaria. CGPA 3.78/4.00, Top 1%</i>	2016 to 2018

RESEARCH AND PROFESSIONAL EXPERIENCE

EMJMD Research Scholar (Mobility Track) <i>Agricultural University of Athens, Greece</i> - Term 3 focus: efficient water use, smart irrigation (30 ECTS). - Continued graduate level research training in smart design and management of irrigation systems. - Accomplished a complete farm level irrigation and crop growth decision support tool, as measured by successful four step simulation (canopy, transpiration, biomass, yield) across five smallholder crops, by engineering NaCROP. Code: GitHub	Sep 2026 to Jan 2027
Summer Intern, Computer Vision Group <i>Dept. of Computer Engineering, ABU, Zaria</i> Accomplished cross condition transfer of crop disease recognition models, as measured by 64.4% classification accuracy (macro AUC 0.882 against a 5.3% chance baseline) across 19 disease classes on 1,543 field images, by extending the Geometry Agnostic Contrastive Learning (GACL) framework with four new geometry invariant neural components. First author paper under review at Biosystems Engineering. Code: GitHub	Jun 2026 to Aug 2026
EMJMD Research Scholar (Mobility Track) <i>UTAD, Portugal</i> - Term 2 focus: precision agriculture for sustainable crop production (30 ECTS). - Accomplished cross system precision agriculture prediction and input cost reduction, as measured by R sq between 0.978 and 0.995 for grape ripening and maize yield regressions and a 2.69 tonne cut in lime demand, by integrating GIS, remote sensing, and ML pipelines across 65+ hectares of Portuguese trial sites. Code: GitHub - Accomplished more reliable soil nutrient and yield mapping, as measured by a roughly 20% reduction in residual kriging uncertainty across 17,864 samples, by developing Multitask Singularity Regression Kriging (MSRK). Code: GitHub - Accomplished predictive harvest date forecasting for grape ripening, as measured by R sq of 0.9784, by building an NDVI and logarithmic regression model at a six hectare UTAD vineyard. Code: GitHub - Accomplished a costed, spatially targeted pest management programme, as measured by a sized pheromone diffuser plan across 11 grape variety blocks, by running terrain characterisation and a 25 trap monitoring network for grape moth across a 42.97 hectare Douro vineyard estate. Code: GitHub	Feb 2026 to Jun 2026
EMJMD Research Scholar (Mobility Track) <i>Public University of Navarre, Spain</i> - Term 1 focus: technological bases for agricultural sustainability (30 ECTS). - Accomplished real time, field deployable crop health screening, as measured by 33.8% balanced accuracy against a 4.55% chance baseline, by building a fourteen module open source pipeline for RGB/multispectral vegetation indexing. Code: GitHub - Accomplished radiometrically consistent multi temporal vegetation monitoring, as measured by a 39% and 31% reduction in temporal NDVI variance, by designing STADOS, a five stage adaptive dark object subtraction framework. Code: GitHub - Accomplished accurate land suitability mapping for livestock slurry application, as measured by RMSE = 7.78 and R sq = 0.77 on a held out set of 127 sites, by building a two layer GIS and regression kriging pipeline near Tudela, Spain. Code: GitHub - Accomplished a 47.8% reduction in annual soil loss, by building a Bayesian grounded digital twin (hold out R sq = 0.90) for gully erosion control, NSE between 0.88 and 0.94. Code: GitHub - Built a six notebook vector/raster GIS and Sentinel 1/2 remote sensing course package in Python and Google Earth Engine. Code: GitHub	Sep 2025 to Jan 2026
Undergraduate Project Assistantships	Mar 2024 to Mar 2025

Depts. of Agricultural and Bioresources Engineering and Animal Science, ABU

- Contributed data collection, statistical analysis, and literature review across concurrent undergraduate research projects spanning water and irrigation systems, soil and erosion assessment, and postharvest and animal systems.

Project Assistant (Remote Data Lead), Smart Greenhouse Automation

2024 to 2025

IAR-ABU, Nigeria

- Automated greenhouse environmental control by engineering adaptive irrigation scheduling algorithms in Python fusing real time soil moisture and microclimate telemetry.

Student Research Assistant, Precision Crop Systems

2022 to 2024

NAERLS, Nigeria

- Trained and validated CNN based crop stress detection models in TensorFlow reaching 78% classification accuracy on multispectral UAV imagery; automated UAV flight path planning.

Assistant Farm Manager (concurrent, part time)

2022 to 2024

Yankari Nasarawa Farms, Nigeria

- Sustained animal welfare and operational efficiency across a commercial poultry facility by managing automated climate control and water dosing systems alongside concurrent research duties.

Project Assistant, Smart Irrigation and Water Systems

2021 to 2022

BOMO Basin Project, Nigeria

- Automated water delivery across a one hectare smart drip irrigation array by implementing closed loop control logic.

Research Intern

2017 to 2020

NCAM, Ilorin and IAR, Zaria, Nigeria

- Benchmarked mechanization efficiency of locally engineered machinery via Python data logging pipelines; supported baseline hydrological research.

PUBLICATIONS

Peer Reviewed

- Halilu, N., Sulaiman, J. (2026). Real Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. IJRSI, 13(7), 2951 to 2989. [DOI](#) | [Code](#)
- Halilu, N., Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. IJRSI, 13(7), 2990 to 3020. [DOI](#)
- Ismail, K., Halilu, N. (2024). AI Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. Proc. AfCPA, 2(1), 100 to 120.
- Halilu, N. (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. IJRSI, 13(7), 2920 to 2950. [DOI](#) | [Code](#)
- Auwal, M., Yusuf, A., Naziru, H. (2020). Physio chemical and bacteriological quality of water for irrigation purpose. Nigerian Research Journal of Engineering and Environmental Sciences, 5(1), 198 to 204.

Submitted and Under Review

- Halilu, N., Sulaiman, J. Geometry Agnostic Contrastive Learning (GACL). Biosystems Engineering, Elsevier. Under review. [DOI](#) | [Code](#)
- Halilu, N., Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. European Journal of Agronomy, Elsevier. Under review. [DOI](#) | [Code](#)
- Halilu, N. (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (*Ipomoea* spp.). Environmental Monitoring and Assessment, Springer Nature. Under review. [DOI](#) | [Code](#)

Preprints

- Halilu, N. (2026). Multitask Singularity Regression Kriging for Spatio Temporal Soil NPK Crop Stress Yield Prediction. Zenodo Preprint. [DOI](#) | [Code](#)
- Halilu, N. (2026). STADOS: Spectral Temporal Adaptive Dark Object Subtraction. Zenodo. [DOI](#) | [Code](#)
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time Series Modelling and Control Systems in Precision Irrigation. Zenodo. [DOI](#)

RESEARCH SOFTWARE AND OPEN DATASETS

- Geospatial Data Analysis: six notebook vector/raster GIS, Sentinel 1/2, and Google Earth Engine curriculum. [DOI](#) | [Code](#)
- Integrated Decision Support Model for Dairy Processing: Ridge regression optimizer, R sq = 0.859 (LOOCV), 5,000 run Monte Carlo simulation. [DOI](#) | [Code](#)
- GIS Based Delineation of Areas Suitable for Livestock Slurry Application: random forest regression kriging (RMSE = 7.78, R sq = 0.77) across 14.35 km sq. [DOI](#) | [Code](#)
- NaCROP (Zaria Crop ET and Irrigation DSS): desktop tool scheduling irrigation for five smallholder crops, water balance closure within 0.1 mm. [Code](#)

TEACHING, LEADERSHIP AND COMMUNITY ENGAGEMENT

Undergraduate Course Tutor

2020 to 2024

ABU Zaria (NAMET), Nigeria

- Ran supplementary tutorials in Differential Equations and Transforms, Irrigation System Design, and Computer Applications in Agric. Engineering.

Secretary General, NAAESB

2023 to 2024

National Assoc. of Agric. Engineering Students

- Directed operational targets and inter chapter scientific meetups for the national student body.

Deputy Coordinator

2020 to 2025

KOJYAF, Zango Shanu, Zaria

- Coordinated tutorial and mentorship programs preparing secondary students for JAMB entrance examinations.

Director, Environmental Protection Agency

2023 to 2024

ABU Students' Union Government

- Directed campus environmental protection and sanitation initiatives.

Physics and Mathematics Teacher

2022 to 2023

Takai Academy, Zaria, Nigeria

- Instructed senior classes in classical physics and computational principles.

TECHNICAL SKILLS

Geospatial AI / GIS:	QGIS, ArcGIS, ENVI, Google Earth Engine, Sentinel imagery, UAV multispectral/hyperspectral imaging
Machine Learning and AI:	PyTorch, TensorFlow, Keras, scikit-learn, CNNs, self supervised and contrastive learning
Crop and Environmental Modelling:	APSIM Next Generation, crop growth and simulation modelling
Climate and Environmental AI:	CMIP6, ERA5, ACCESS-S2, climate downscaling and forecasting
3D Computer Vision:	COLMAP, Open3D, Multi View Stereo, Pix4D, Agisoft Metashape
Sensor Analytics:	Time series fusion of soil moisture, microclimate, and environmental telemetry
Programming:	Python, MATLAB, R, Pandas, NumPy, SciPy, Jupyter, Git/GitHub, Zenodo

CONFERENCES, SEMINARS AND WORKSHOPS

Digital Agroecology Cluster Conference (EU funded), Brussels (2025); Vibrational Spectroscopy and Chemometrics in Agro Food Systems (EU funded) (2025); IEEE International Conference on Smart Agriculture Technologies (AgriTech), Valencia, Spain (2024); 6th African Regional Conference on Irrigation and Drainage (ARCID) (2025); Mechanistic Modeling of Rainfall Driven Soil Solute Transport, University of Florida (2025); NAERLS Technical Seminar Series (2023).

AWARDS AND HONORS

Erasmus Mundus Joint Master Scholarship, European Union (2025); Best Project Award, ABU Senate (2024); Best Graduating Student, Agricultural and Bioresources Engineering (2024); Nigerian Federal Government Award for Academic Excellence (2024); Best Undergraduate Research Paper Award, ABU (2024); Best 300 Level Student, Dept. of Agric. and Bioresources Engineering, ABU (2022); Best 200 Level Student, Dept. of Agric. and Bioresources Engineering, ABU (2021); Award of Merit as Tutor, Faculty of Engineering, ABU (2021); Best Graduating Student, National Diploma, Samaru College of Agriculture (2019); PTFDF Undergraduate Fellow (2023); Needs Based University Fellowship (2022).

PROFESSIONAL MEMBERSHIPS

American Society of Agricultural and Biological Engineers (ASABE); Nigerian Institution of Agricultural Engineers (NIAE); Nigerian Society of Engineers (NSE); Pupil Engineer, COREN.

REFEREES

Prof. Miguel Ángel Campo Bescós, Public University of Navarre, miguel.campo@unavarra.es, +34 948 16 9077

Prof. José Tadeu Marques Aranha, UTAD, Portugal, j_aranha@utad.pt, ext. 4493

Dr. Ibrahim Babangida Dalha, Ahmadu Bello University, ibdalha@abu.edu.ng, +234 806 687 3858